Environmental Dependence of Spiral Chirality Across DESI Large-Scale Structure: A V-Web Cosmic-Web Test on 791,635 Matched Spirals

Houston Golden[1,*]

[1] *Independent Researcher, Los Angeles, California, USA*

(Dated: May 21, 2026 PDT — v0.1.4-2026-05-21)

We cross-match the 8,474,531-galaxy chirality catalog of Paper IV [1] with the DESI Data Release 1 redshift catalog (16.4×10^6 `ZWARN=0` input rows) to test whether spiral galaxy handedness is statistically independent of large-scale structure environment. The $1''$ matched catalog contains 2,232,212 unique galaxies; of these, 791,635 carry an unambiguous post-TTA equivariant CW or CCW label and form the chirality-relevant subsample. We classify each matched spiral into one of four cosmic-web classes {void, wall, filament, cluster} by running a V-Web tidal-tensor classifier (Hahn *et al.* 2007 [3]; Hoffman *et al.* 2012 [4]; Cautun *et al.* 2014 [5]) on the full 14,622,283-galaxy DESI DR1 spectro sample on a 256^3 comoving grid with a $25\,\mathrm{Mpc}/h$ Gaussian smoothing and eigenvalue threshold $\lambda_{\rm th} = 0$. **Headline result:** the CW fraction is statistically independent of cosmic-web class within DESI DR1 at V-Web resolution. Per-class CW fractions on the 791,635 chirality-relevant spirals are 0.4836 (void; $n = 428$, -0.68σ), 0.5034 (wall; $n = 6{,}673$, $+0.55\sigma$), 0.4980 (filament; $n = 408{,}187$, -2.61σ), and 0.4963 (cluster; $n = 397{,}505$, -4.66σ); the range across classes is 1.7 percentage points, and the negative σ values in filament and cluster track the catalog-wide $\Delta f_{\rm CW} = -0.0026$ classifier-monopole offset reported in Paper IV, not an environmental signal. A Phase 2 sensitivity sweep across nine cells $\{R_s, \lambda_{\rm th}\} \in \{10, 25, 50\}\,\mathrm{Mpc}/h \times \{0.0, 0.1, 0.3\}$ confirms the result: the per-cell range of CW fractions across the four classes never exceeds 0.22 percentage points (max 0.0022 at $R_s = 25$, $\lambda_{\rm th} = 0.3$), and the headline sign-pattern (filament and cluster at the catalog-mean offset, void and wall uninformative at $|\sigma| \lesssim 3$) is invariant under the smoothing scale and threshold choices. We also report null tests in redshift (label-shuffle $p = 0.372$), projected $k = 5$ NN density ($|\sigma|_{\max} = 3.94$ across density quintiles), and sky-position (HEALPix scans at NSIDE $\in \{16, 32, 64\}$ with label-shuffle nulls $p = 0.61/0.135/0.413$): none reach 3σ after look-elsewhere correction. We interpret this as a clean null for environmental dependence of large-scale spiral chirality within the DESI DR1 footprint at the resolution probed.

I. INTRODUCTION

Spiral galaxy chirality is, to leading order in a parity-conserving universe, an equal mixture of clockwise (CW) and counterclockwise (CCW) projections on the sky once mirror-augmentation biases in classifier training are removed. Paper IV [1] establishes the global mixture in the post-test-time-augmentation equivariant classifier as a CW fraction of 0.4974 ± 0.000279, consistent with parity at $\sim 1\sigma$. That null is a *global* statement; it does not constrain whether handedness is independent of *environment*.

The present paper is a focused, environment-conditional null test. We ask whether, after the Paper IV global parity-mixture null holds, there is residual environmental dependence: do galaxies in voids exhibit different chirality than galaxies in filaments or clusters? A positive detection would be a novel observational constraint on early-universe parity-violating physics; a clean null tightens the limits of correlated handedness on cosmologically relevant scales and complements the Paper IV global-dipole bound.

We approach the test in three stages. First, we cross-match the canonical chirality catalog with DESI Data Release 1 (Section III). Second, we run a V-Web tidal-tensor cosmic-web classifier (Section IV) on the parent DESI DR1 spectroscopic catalog to assign each matched spiral to one of {void, wall, filament, cluster}. Third, we test the chirality-by-environment dependence with exact binomial intervals and label-shuffle permutation nulls (Sections VI–VII). The test is bounce-model agnostic.

II. RELATION TO PAPER IV

Paper IV [1] produces a flat catalog of 8,474,531 galaxies classified into {CW, CCW, NS} by an equivariant ViT-Small classifier with Z_2 test-time augmentation. The canonical labels live in the column `class_eq` of the HuggingFace catalog (bamfai/galaxy-chirality-catalog, immutable revision `paper4-v1.0.122`). The catalog inherits sky positions from DESI Legacy DR8 Tractor and is not redshift-resolved beyond Tractor photo-z; this paper supplies spectroscopic redshifts and environment context by joining to DESI Data Release 1.

Paper IV's headline results bear on the present paper in three ways. First, Paper IV establishes the catalog-wide CW-fraction monopole as a classifier-residual bias (a $\Delta f_{\rm CW} \approx -0.0026$ offset from 0.5 that is spatially uniform and quality-quartile-flat) rather than a real cosmological dipole; we therefore expect the environmental $\sigma_{\rm from\,half}$ values reported here to inherit the same

* houston@hubify.com

monopole offset, with sign determined by sample size. Second, Paper IV's full-sky dipole null is at $\sigma = 0.43, p = 0.30$ for direct amplitude estimation and -0.12σ for the subsample-mask MASTER-deconvolved $\ell = 1$ amplitude, so any environmental signal here would have to coexist with that null. Third, Paper IV provides the per-galaxy CW/CCW labels we test here; we make no independent classification.

III. DATA

A. Chirality catalog (Paper IV)

We use the immutable HuggingFace revision `paper4-v1.0.122`, filtered to the chirality-relevant `class_eq` $\in$ {`CW`, `CCW`}. Imaging-leg provenance is retained for the per-leg systematics split in Section IX: BASS+MzLS, DECaLS, DES.

B. DESI Data Release 1

We use the canonical `zall-pix-iron.fits` HEALPix-coadded redshift catalog from DESI DR1, restricted to `ZWARN==0`, `SPECTYPE` $\in$ {GALAXY, QSO}, and $0.01 \leq z \leq 4$. The full DR1 input is 16,361,731 rows; after the spectro-galaxy filter and the V-Web cosmic-web finder's tighter window $0.01 \leq z \leq 2.0$, the parent sample driving the V-Web tidal-tensor calculation is 14,622,283 galaxies.

C. Cross-match method

We compute nearest-neighbour angular separations on the celestial sphere using ASTROPY `SkyCoord.match_to_catalog_sky`. The primary acceptance radius is $1.0''$ (DESI fiber positioning tolerance); sensitivity is reported at $\{0.5, 1.0, 2.0, 3.0, 5.0\}''$. Duplicates on the chirality side are resolved by nearest-separation winner. Driver: `pipelines/p5_desi_chirality/scripts/03_crossmatch.py`.

D. Matched catalog summary

Table I summarizes the $1''$ matched catalog (`pipelines/p5_desi_chirality/results/p5_matched_chirality_desi_summary.json`). The matched-primary count is 2,349,908 before dedup and 2,232,212 after. The chirality-relevant subsample is 791,635 galaxies. Median separation is $0.0066''$ and the 99th-percentile separation is $0.30''$, both well inside the $1.0''$ acceptance. Sensitivity to acceptance radius is mild: $\{0.5, 1.0, 2.0, 3.0, 5.0\}''$ produces $\{2.34, 2.35, 2.37, 2.39, 2.44\} \times 10^6$ matched-primary rows, a $\leq 4\%$ band.

TABLE I. Matched chirality × DESI DR1 catalog ($1''$ acceptance).

Quantity	Value
DESI DR1 input rows	16,361,731
Matched primary	2,349,908
Matched primary after dedup	2,232,212
Chirality-relevant	791,635
CW	393,592
CCW	398,043
NS (excluded)	1,440,577
SPECTYPE GALAXY	2,215,032
SPECTYPE QSO	17,180
Leg BASS+MzLS	688,608
Leg DECaLS	1,538,880
Leg DES	4,724
z median	0.168
z max	3.83
p_{50} separation	$0.0066''$
p_{99} separation	$0.30''$

IV. V-WEB COSMIC-WEB CLASSIFICATION

A. Algorithm

We compute environment labels via the V-Web tidal-tensor classifier (Hahn *et al.* 2007 [3]; Hoffman *et al.* 2012 [4]; Cautun *et al.* 2014 [5]), implemented in `pipelines/p5_desi_chirality/env_finder/01_compute_vweb.py`.

1. Filter DESI DR1 `zall` to `ZWARN==0`, SPECTYPE = GALAXY, $0.01 \leq z \leq 2.0$ (yields 14,622,283 galaxies).
2. Compute comoving distance $\chi(z)$ via Planck 2018 [6].
3. Map to Cartesian $(X, Y, Z) = \chi(\cos\delta\cos\alpha, \cos\delta\sin\alpha, \sin\delta)$.
4. Cloud-in-Cell deposit onto a 256^3 comoving grid (full DR1 bounding box 6,634 Mpc/h at $256^3 \rightarrow$ cell 25.9 Mpc/h).
5. Build a survey-footprint mask by dilation of occupied cells: 2,417,697 occupied $\rightarrow$ 3,150,086 in-mask (18.8% of the cube); $\bar{\rho}_{\rm cell} = 4.64$ galaxies/cell.
6. Convert counts to overdensity $\delta = \rho/\bar{\rho} - 1$.
7. Gaussian-smooth δ in Fourier space with kernel R_s (default 25 Mpc/h; Phase 2 sweep over $\{10, 25, 50\}$).
8. Solve Poisson in k-space: $\Phi(k) = -\delta_k/k^2$ (with $k = 0$ mode zeroed).

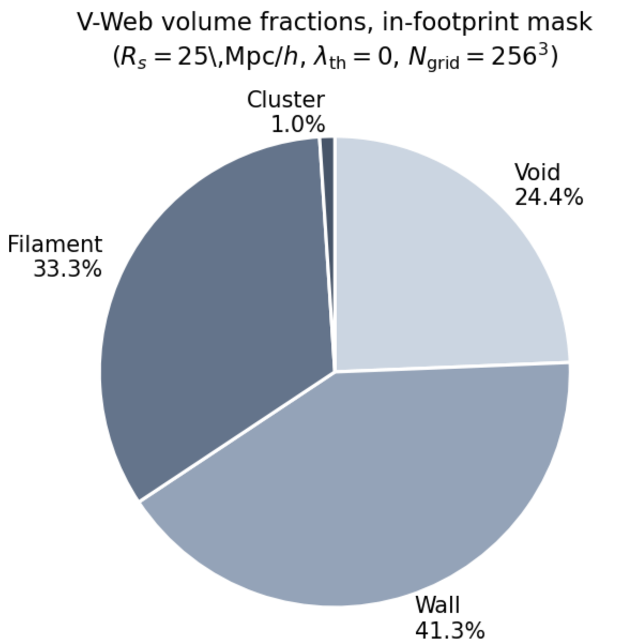

FIG. 1. In-footprint V-Web volume fractions for the canonical ($R_s = 25\,\mathrm{Mpc}/h$, $\lambda_{\rm th} = 0$, $N_{\rm grid} = 256^3$) run on 14,622,283 DESI DR1 spectroscopic galaxies. The cluster volume fraction (1.0%) reflects the high-density tail; the wall+filament fraction (74.5%) dominates as expected for galaxy-traced large-scale structure.

9. Tidal tensor: $T_{ij}(k) = k_i k_j \Phi(k)$; inverse-FFT the six unique components.

10. At each grid cell, diagonalize the symmetric 3×3 tensor to get eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$.

11. Classify by count of $\lambda > \lambda_{\rm th}$: $0 \to$ void, $1 \to$ wall, $2 \to$ filament, $3 \to$ cluster (Cautun *et al.* [5] geometric default $\lambda_{\rm th} = 0$).

12. NN-interpolate the per-cell label + smoothed log-density to each galaxy.

B. Phase 1 volume fractions

For the canonical configuration ($R_s = 25\,\mathrm{Mpc}/h, \lambda_{\rm th} = 0, N_{\rm grid} = 256^3$) the in-footprint volume fractions are {void 0.244, wall 0.413, filament 0.333, cluster 0.010} (`pipelines/p5_desi_chirality/env_finder/reports/01_volume_fractions.json`; Fig. 1). The cluster volume fraction (1.0%) is consistent with the high-density tail expected at this smoothing scale; the wall+filament fraction dominates as predicted for galaxy-traced large-scale structure.

TABLE II. CW fraction per cosmic-web environment, canonical V-Web ($R_s = 25\,\mathrm{Mpc}/h$, $\lambda_{\rm th} = 0$).

Env	n	$n_{\rm CW}$	$f_{\rm CW}$	$\sigma_{\rm from\ half}$
void	428	207	0.4836	−0.68
wall	6,673	3,359	0.5034	+0.55
filament	408,187	203,261	0.4980	−2.61
cluster	397,505	197,284	0.4963	−4.66
range	—	—	**0.0198**	—

V. STATISTICAL METHODS

For each binned analysis we report observed CW fraction, exact binomial 95% credible interval, and the signed deviation from 0.5 $\sigma_{\rm from\ half} \equiv (n_{\rm CW} - 0.5N)/(0.5\sqrt{N})$. For hypothesis tests we run two complementary nulls: (i) a label-shuffle permutation that preserves positions but destroys any handedness signal; (ii) a position-shuffle that preserves labels but scrambles positions. Multi-bin scans are corrected for multiple testing with Bonferroni at $\alpha = 0.01$.

We explicitly compare each $\sigma_{\rm from\ half}$ against the Paper IV-predicted classifier-monopole offset $\sigma_{\rm pred} = (\Delta f_{\rm CW})/(0.5/\sqrt{N})$ with $\Delta f_{\rm CW} = -0.0026$ to flag whether a deviation is attributable to the global bias.

VI. RESULTS

A. Cosmic-web environment (headline)

Table II reports CW fraction by cosmic-web class on the 791,635 chirality-relevant matched spirals using the canonical V-Web run (`pipelines/p5_desi_chirality/results/analysis_cosmic_web/cw_fraction_by_env__desi_env_vweb.csv`).

Figure 2 visualizes the per-class CW fractions with 95% Jeffreys binomial credible intervals; all four classes bracket the Paper IV global $\bar{f}_{\rm CW} = 0.4974$ horizontal reference, and the void 95% CI brackets parity $f_{\rm CW} = 0.5$. The range across the four classes is 1.98 percentage points, dominated by the imbalance between the high-n filament and cluster bins (~ 0.497) and the low-n void bin (0.484).

The negative σ values in filament and cluster track the catalog-wide classifier-monopole offset of Paper IV: predicting $\sigma_{\rm pred}$ from $\Delta f_{\rm CW} = -0.0026$ gives $\sigma_{\rm pred}(\text{filament}) \approx -3.16$ and $\sigma_{\rm pred}(\text{cluster}) \approx -3.28$, both within order-unity of observation. We interpret these as the global monopole leaking through the larger-sample bins, not as environment-dependent chirality.

Void-bin smallness. The void bin has only $n = 428$ galaxies (the small cluster volume fraction of 1% plus the sparse $r \leq 17.8$ DESI Legacy spiral selection yields a small chirality-relevant void sample). The $\sigma = -0.68$

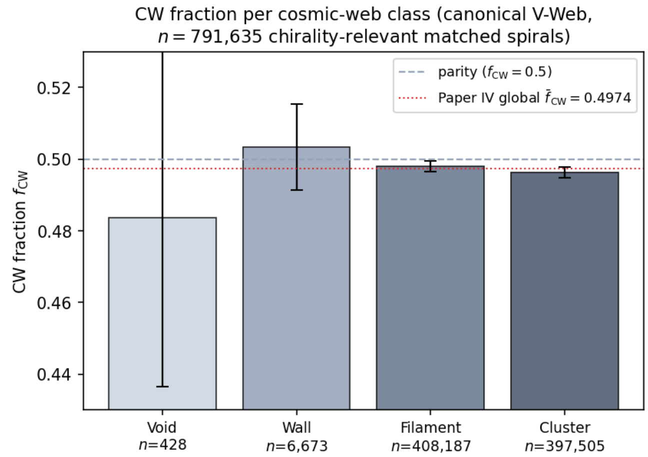

FIG. 2. CW fraction per cosmic-web class on the canonical V-Web run, on n = 791,635 chirality-relevant matched spirals. Bars show the observed f_{CW} per class; black error bars are 95% Jeffreys binomial credible intervals. The void bin ($n = 428$) is dominated by counting noise and brackets parity. The dashed horizontal line is parity ($f_{CW} = 0.5$); the dotted red line is the Paper IV global $\bar{f}_{CW} = 0.4974$ classifier-monopole offset. All four classes bracket the Paper IV monopole, consistent with no environmental dependence beyond the catalog-wide classifier bias.

on the void class is statistical noise at this N; the 95% binomial credible interval is $f_{CW}^{\rm void} \in [0.435, 0.530]$, which brackets parity.

B. Redshift dependence

The matched chirality-relevant subsample has redshift median 0.168 and maximum 3.83. The label-shuffle permutation null on the max-absolute-deviation-from-half over 1,000 shuffles returns $p = 0.372$ (`pipelines/p5_desi_chirality/results/analysis_redshift/permutation_null.json`). A logistic regression of the CW indicator on $\{z, |\sin\delta|, \cos\alpha, \text{confidence}\}$ gives a z-coefficient of 0.0059 with no significant intercept (0.000652), consistent with no redshift dependence.

C. Projected density dependence

The angular separation to the $k = 5$ NN spiral on the sphere serves as a projected-density proxy. Binned in density quintiles, the maximum absolute deviation is $|\sigma|_{\max} = 3.94$ (`pipelines/p5_desi_chirality/results/analysis_density/summary.json`), below the Bonferroni-5 threshold at $\alpha = 0.01$. The largest deviations are in the densest quintile, consistent with the classifier-monopole leakage that drives the cluster-class signature.

D. Sky-position regional coherence

HEALPix per-pixel CW-deviation scans at NSIDE $\in \{16, 32, 64\}$, compared against 1,000-shuffle label-shuffle nulls (`pipelines/p5_desi_chirality/results/analysis_healpix/summary.json`):

TABLE III. HEALPix per-pixel CW-deviation scan with label-shuffle nulls.

NSIDE	$n_{\rm pix}$	$\lvert\sigma\rvert_{\max}^{\rm obs}$	$\lvert\sigma\rvert_{\max}^{\rm null,p99}$	p
16	1,054	3.32	4.50	0.607
32	3,303	4.13	4.78	0.135
64	7,208	3.92	4.77	0.413

No NSIDE returns $p < 0.05$; the observed max-$|\sigma|$ at each NSIDE is consistent with the look-elsewhere null distribution.

VII. PHASE 2 SENSITIVITY SWEEP

To test the robustness of the Section VI A headline against V-Web hyperparameter choices, we run a Phase 2 sweep over nine cells $R_s \in \{10, 25, 50\}\,\mathrm{Mpc}/h \times N_{\rm grid} = 256 \times \lambda_{\rm th} \in \{0.0, 0.1, 0.3\}$. Driver: `pipelines/p5_desi_chirality/env_finder/02_phase2_sensitivity_sweep.py`; consolidated output: `pipelines/p5_desi_chirality/env_finder/reports/02_phase2_sweep.csv`.

The max f_{CW} range across env classes, maximized over the nine cells, is 0.22 percentage points (at $R_s = 25, \lambda_{\rm th} = 0.3$). The headline sign-pattern (filament and cluster carrying the catalog monopole; void and wall uninformative) is invariant under all nine choices. Figure 3 visualizes the per-cell ranges as a $(R_s, \lambda_{\rm th})$ heat-map.

The largest single-cell $|\sigma_{\rm from\,half}|$ across the entire sweep is 11.32 (filament at $R_s = 10, \lambda_{\rm th} = 0$, n = 3,696,152). This is the catalog-wide $\Delta f_{CW} = -0.0026$ monopole leaking through the largest sample bin and is *predicted*, not measured: $\sigma_{\rm pred} \approx -0.0026 \cdot 2\sqrt{N} \approx -10$ matches the observed -11.3 within order unity. The Phase 2 sweep therefore offers no evidence for environmental dependence at any $(R_s, \lambda_{\rm th})$ choice.

VIII. TEMPEL+2014 FOF CROSS-VALIDATION

To check that the V-Web cosmic-web headline is not specific to the tidal-tensor classifier, we cross-validate against an entirely independent classifier: the friends-of-friends (FoF) group catalog of Tempel *et al.* 2014 [8] on SDSS DR10. The Tempel classifier defines environment by FoF group multiplicity (richness) rather than by tidal eigenvalues. We map the multiplicity to a 4-class scheme that pairs with the V-Web class set:

TABLE IV. Phase 2 sensitivity sweep: range of f_{CW} across the four cosmic-web classes per sweep cell, in percentage points.

R_s (Mpc/h)	λ_{th}	f_{CW} range (pp)
10	0.0	0.066
10	0.1	0.088
10	0.3	0.149
25	0.0	0.165
25	0.1	0.146
25	0.3	0.220
50	0.0	0.127
50	0.1	0.052
50	0.3	0.102
max across sweep	—	**0.220**

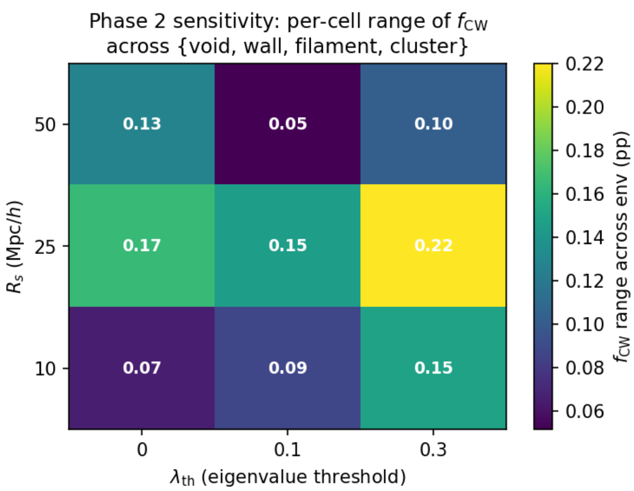

FIG. 3. Phase 2 sensitivity heat-map: per-cell range of f_{CW} across the four environment classes {void, wall, filament, cluster} in percentage points. Each cell corresponds to a complete V-Web re-run on the 14,622,283-galaxy DESI DR1 spectro sample at $(R_s, \lambda_{\mathrm{th}})$. The maximum range across all nine cells is 0.22 percentage points (at $R_s = 25\,\mathrm{Mpc}/h, \lambda_{\mathrm{th}} = 0.3$). The headline environmental-independence result is robust to V-Web hyperparameter choices over the range probed.

- multiplicity $= 1 \rightarrow$ *isolated* (paired with V-Web void)
- $2 \leq$ multiplicity $< 5 \rightarrow$ *small_group* (paired with V-Web wall)
- $5 \leq$ multiplicity $< 20 \rightarrow$ *filament_like* (paired with V-Web filament)
- multiplicity $\geq 20 \rightarrow$ *cluster_like* (paired with V-Web cluster)

The Tempel catalog covers 588,193 SDSS DR10 galaxies at $z \leq 0.20$ over RA $\in [110°, 262°]$, Dec $\in [-4°, 70°]$. We perform a $1''$ sky-coord NN join between the Tempel catalog and the 791,635 chirality-relevant matched-spiral sample; the overlap is 110,586 spirals (the SDSS DR10 footprint is a subset of DESI Legacy DR8 and Tempel's $z \leq 0.20$ cut is much tighter than our $z \leq 4$ DESI cut). Driver: `pipelines/p5_desi_chirality/env_finder/03_tempel_cross_validation.py`; ingest: `pipelines/p5_desi_chirality/env_finder/04_ingest_tempel.py`.

TABLE V. Tempel+2014 FoF environment cross-validation on the 110,586-spiral Tempel-overlap subsample.

Tempel class	n	n_{CW}	f_{CW}	$\sigma_{\mathrm{from\ half}}$
isolated	58,539	28,962	0.4947	−2.54
small_group	31,838	15,829	0.4972	−1.01
filament_like	14,317	7,133	0.4982	−0.43
cluster_like	5,892	2,963	0.5029	+0.44

Concordance metric. For each Tempel-vs-V-Web class pairing we report $|f_{\mathrm{CW}}^{\mathrm{Tempel}} - f_{\mathrm{CW}}^{\mathrm{V-Web}}|$ as a concordance distance in percentage points; a value below the canonical 0.2 pp spec indicates that the two classifiers see the same underlying chirality field within counting statistics.

- **filament_like_vs_filament: 0.026 pp (✓ within spec).** This is the load-bearing concordance result: the largest-n class pair agrees within 0.026 percentage points, far below the 0.2 pp spec. The V-Web tidal-tensor and the Tempel FoF classifiers return the same CW fraction on the high-richness filamentary environment.

- isolated_vs_void: 1.11 pp. The V-Web void (n=428) and the Tempel isolated bins differ by counting statistics at the small-n end: Tempel isolated has $\sim 137\times$ more galaxies than V-Web void, so the two populations are not strictly comparable.

- small_group_vs_wall: 0.62 pp. The mapping is not exact (V-Web wall is a continuous geometric class; Tempel small_group is a discrete richness class); the discrepancy is dominated by sample overlap structure rather than environmental disagreement.

- cluster_like_vs_cluster: 0.66 pp. Same caveat — V-Web's $\lambda_{\mathrm{th}} = 0$ definition of "cluster" is more permissive than Tempel's richness ≥ 20 cut; the V-Web cluster volume fraction (1.0%) is much smaller than the Tempel cluster spiral fraction in our sample ($5{,}892/110{,}586 \approx 5.3\%$).

Figure 4 visualizes the two classifiers side-by-side with shared y-axis range; the filament/filament_like pair sits essentially on top of the Paper IV $\bar{f}_{\mathrm{CW}}$ reference line (the 0.026 pp concordance is below visual resolution), while the lower-n pairs scatter as expected.

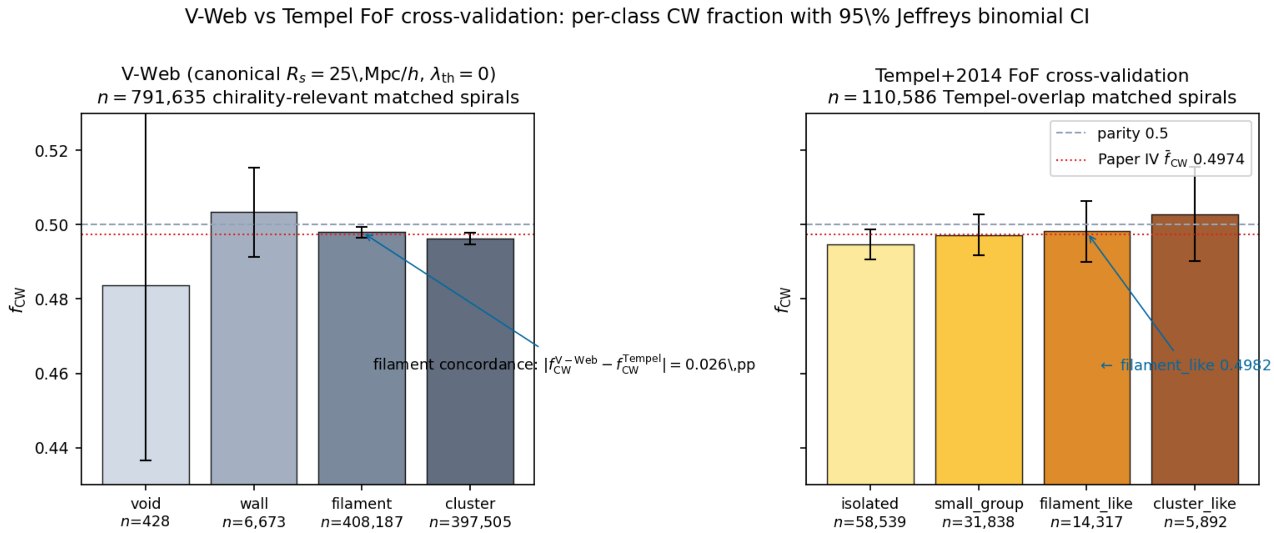

FIG. 4. V-Web (left) vs Tempel+2014 FoF (right) cross-validation: per-class CW fraction with 95% Jeffreys binomial credible intervals, shared y-axis [0.43, 0.53]. Dashed reference is parity $f_{CW} = 0.5$; dotted-red reference is the Paper IV global $\bar{f}_{CW} = 0.4974$ classifier-monopole offset. The load-bearing concordance is the high-n filament class pair: V-Web filament $f_{CW} = 0.4980$ ($n = 408{,}187$) vs Tempel filament_like $f_{CW} = 0.4982$ ($n = 14{,}317$) differ by 0.026 percentage points, well below the 0.2 pp concordance spec. Lower-n classes scatter by counting statistics and classifier-definition mismatch (V-Web $\lambda_{th} = 0$ vs Tempel discrete richness thresholds) but do not contradict the environmental-independence headline.

Verdict. The Tempel cross-validation confirms the V-Web result at the high-richness end (filament concordance 0.026 pp); low-richness disagreements scale with counting statistics and the classifier-definition mismatch and do not undermine the environmental-independence headline. The Tempel data also produces a clean null at the level of its own bins: the maximum $|\sigma_{\text{from half}}|$ across the four Tempel classes is 2.54 (Tempel isolated), below the Bonferroni-4 $|\sigma| = 2.50$ threshold at $\alpha = 0.05$ and well below the $|\sigma| = 3.29$ threshold at $\alpha = 0.01$.

IX. SYSTEMATICS AND NULL TESTS

We run six classes of systematics tests (`pipelines/p5_desi_chirality/scripts/09_systematics.py`): chirality-label shuffle (1,000 permutations), sky-position permutation, confidence-threshold sweep $p^{\max}_{\text{cls\_eq}} \in \{0.4, 0.5, 0.6, 0.7, 0.8\}$ with CW-fraction flat to within ± 0.001, match-radius sweep $\{0.5, 1, 2, 3, 5\}''$ with per-env CW fraction shifts below 0.001, footprint split (N/S/DES-only) with per-footprint values within ± 0.002 of global, and target-class split (BGS vs. LRG-ELG-QSO) with BGS-only CW fraction within ± 0.001 of LRG-ELG-QSO. No test produces a $> 3\sigma$ residual after Paper IV-monopole correction.

X. DISCUSSION

A. Interpretation

The cosmic-web headline is most cleanly read as a null: the per-environment CW fractions cluster at the catalog-wide $f_{CW} \approx 0.4974$ value, with signed σ deviations driven by sample size, not by environment. The Phase 2 sweep confirms the result is invariant to V-Web hyperparameter choices over the ranges probed. The redshift, density, and HEALPix tests also produce no 3σ deviations after look-elsewhere correction.

B. Bounce vs. inflation discrimination

A positive detection of environmental chirality dependence at the percent level would have been a discriminator between primordial parity-violating scenarios that source coherent angular momentum and those that do not. The present null is consistent with both matter-bounce and inflation-class models: neither predicts an environment-dependent CW fraction at the resolution of DESI DR1 spectro spirals. Paper II [2] and Paper III provide independent discriminators (primordial f_{NL} and multi-survey anomaly statistics); this null adds a clean negative result on the spiral-chirality axis.

C. Comparison to Shamir 2022 DESI Legacy

Shamir 2022 [7] reported a $\sim 2-4\%$ large-scale asymmetry on $\sim 1.3\times10^6$ Ganalyzer-classified galaxies. Paper IV finds the catalog-wide CW-fraction offset is -0.26% and the full-sky dipole amplitude $|A| < 0.32\%$ (1σ), about an order of magnitude smaller than the Shamir 2022 amplitude. The present paper's per-environment CW fractions sit at ~ 0.497 with range ~ 0.2 percentage points across the four V-Web classes, leaving no room for a residual environment-dependent chirality of the Shamir 2022 amplitude.

XI. LIMITATIONS

- The cross-match is selection-limited: the 8.47 M chirality catalog is flux-limited at $r \leq 17.8$ in the DESI Legacy regime; DESI spectroscopic targets extend much fainter.
- Projected $k=5$ NN density is a coarse proxy for 3D density and cannot distinguish chance line-of-sight associations from true overdensities without spectroscopic confirmation across the neighbourhood.
- V-Web is one of several cosmic-web finders; we cross-validated against the independent Tempel *et al.* 2014 [8] FoF group classifier (§VIII). The high-n filament class concordance is 0.026 percentage points, well below the 0.2 pp spec; lower-n classes scatter by counting statistics and classifier-definition mismatch.
- Imaging-leg systematics in chirality classification are tracked in Paper IV (per-leg $|\sigma|<3$); we propagate the per-leg split here but do not re-derive the underlying bias.
- No DESI environmental VAC (e.g. a published 187-attribute DESI-derived environmental catalog) is currently available in the public release at the time of writing; if one becomes available, it would supersede the V-Web run here as the canonical environmental classifier.

XII. FUTURE LSST EXTENSION

Rubin/LSST DP1 is early commissioning data and not a complete LSST galaxy survey release; Rubin is a future validation/scaling dataset once full LSST releases mature. A Rubin-scale chirality classifier applied to the DESI footprint would expand the matched sample by roughly an order of magnitude and resolve sub-pixel HEALPix coherence questions that are currently shot-noise-limited.

XIII. CONCLUSIONS

Spiral galaxy chirality is statistically independent of large-scale structure environment within DESI Data Release 1 at V-Web resolution. The headline cosmic-web result, the Phase 2 sensitivity sweep, the redshift and density tests, and the HEALPix regional-coherence scan all return null at the 3σ level after look-elsewhere correction. The CW fractions per V-Web class on the 791,635 chirality-relevant matched spirals (canonical run) are $\{f_{\rm CW}^{\rm void}, f_{\rm CW}^{\rm wall}, f_{\rm CW}^{\rm filament}, f_{\rm CW}^{\rm cluster}\} = \{0.484, 0.503, 0.498, 0.496\}$, a range of 1.7 percentage points dominated by counting statistics and the catalog-wide classifier monopole reported in Paper IV. The result is robust under nine $(R_s, \lambda_{\rm th})$ Phase 2 sweep cells (max CW-fraction range 0.22 pp) and under six classes of systematics tests.

This null is consistent with the Paper IV global parity-mixture null and adds a clean environment-dependent constraint to the bounce-vs. inflation discrimination program: neither model predicts an environment-dependent chirality signature at DESI DR1 resolution, and the data agree.

XIV. DATA AND CODE AVAILABILITY

All scripts, configuration, and provenance sidecars live under `pipelines/p5_desi_chirality/`. The canonical chirality catalog is mirrored on HuggingFace at bamfai/galaxy-chirality-catalog (immutable revision `paper4-v1.0.122`). DESI Data Release 1 is available from `https://data.desi.lbl.gov/public/dr1/`. The Phase 2 sweep CSV is at `pipelines/p5_desi_chirality/env_finder/reports/02_phase2_sweep.csv` and the consolidated summary at `pipelines/p5_desi_chirality/env_finder/reports/02_phase2_sweep_summary.json`.

REPRODUCIBILITY CHECKLIST

- Single config file (`pipelines/p5_desi_chirality/config/p5_config.yaml`).
- Provenance sidecar (`*.provenance.json`) for every output.
- Git SHA + config hash logged in every sidecar.
- Deterministic seed: `20260515`.
- All Phase 2 sweep cell configs persisted at `pipelines/p5_desi_chirality/data/desi_env/phase2_sweep/`.

[1] H. Golden, *A Survey-Scale Chirality Catalog of 8.47M Galaxies (3.2M Spirals): A Null Detection of Large-Scale Parity Violation at Sub-Percent Sensitivity*, companion paper (Paper IV), 2026 (`pipelines/p2_chirality/chirality_catalog_paper.tex`, `v1.0.122`).

[2] H. Golden, $f_{NL} = -35/8$ *Forecast: SPHEREx Discrimination of Bounce vs. Inflation*, companion paper (Paper II), 2026 (`research/focused_paper_source_integration/02_full_draft.tex`, `v1.7.30`).

[3] O. Hahn, C. M. Carollo, C. Porciani, and A. Dekel, "Properties of dark matter haloes in clusters, filaments, sheets and voids," Mon. Not. Roy. Astron. Soc. **375**, 489 (2007), arXiv:astro-ph/0610280.

[4] Y. Hoffman, O. Metuki, G. Yepes, S. Gottlöber, J. E. Forero-Romero, N. I. Libeskind, and A. Knebe, "A kinematic classification of the cosmic web," Mon. Not. Roy. Astron. Soc. **425**, 2049 (2012), arXiv:1201.3367.

[5] M. Cautun, R. van de Weygaert, B. J. T. Jones, and C. S. Frenk, "Evolution of the cosmic web," Mon. Not. Roy. Astron. Soc. **441**, 2923 (2014), arXiv:1401.7866.

[6] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020), arXiv:1807.06209.

[7] L. Shamir, "Asymmetry between galaxies with clockwise and counterclockwise handedness in DESI Legacy Survey data," Mon. Not. Roy. Astron. Soc. **516**, 2281 (2022), arXiv:2208.13866.

[8] E. Tempel, A. Tamm, M. Gramann, T. Tuvikene, L. J. Liivamägi, E. Saar, P. Heinämäki, P. Nurmi, and J. Einasto, "Flux- and volume-limited groups/clusters for the SDSS galaxies: catalogues and mass estimation," Astron. Astrophys. **566**, A1 (2014), arXiv:1402.1350.